

Competitive Multi-Robot Parking Game: Vision-Based Localization and Gesture-Based Remote Control

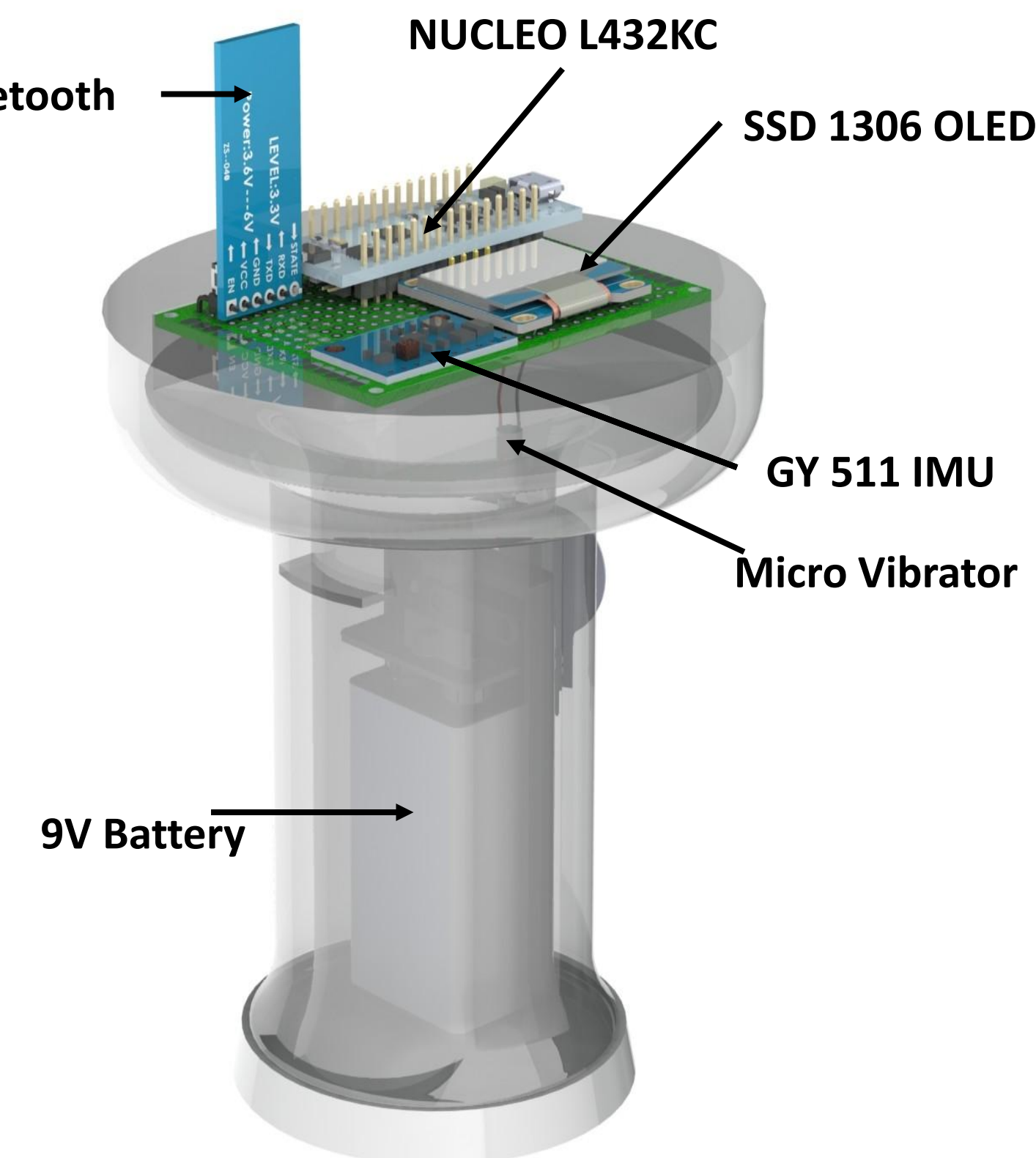
EECS 373: Introduction to Embedded System Design | Guanyu Xu, Haobo Fang, Varun Agrawal, and Xiang Jiang

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Introduction

- A fast paced, interactive parking game using four Zumo robots on a 4x4 ft LED marked field.
- The system uses STM32 microcontrollers, **Vision-Based localization, and Gesture-Based remote control** to drive gameplay.
- The platform highlights **embedded integration, real time multi-agent coordination, and dynamic visual and haptic feedbacks**.
- Robust, modular system architecture allows for further expansion.

Remote Controller Design



GAME ON!

- Players:** 4 Zumo cars, 4 Colors, 4 Starting Positions
- Rounds:** 5 Total rounds, Pick the highest total scores
- Parking Spots:** 10 randomly generated per round, **available spots** will be indicated by **White LED** lights
- Time Limit:** **30 seconds** per round
- Between Rounds:** Zumo robots are free to move across rounds
- Scoring Rule:** A robot scores only if it parks in a spot that turns into its **assigned color** during the round
- Warning:** Participants are strictly prohibited from touching the robots, the arena, or any game elements during the match

Methodology

Playground:

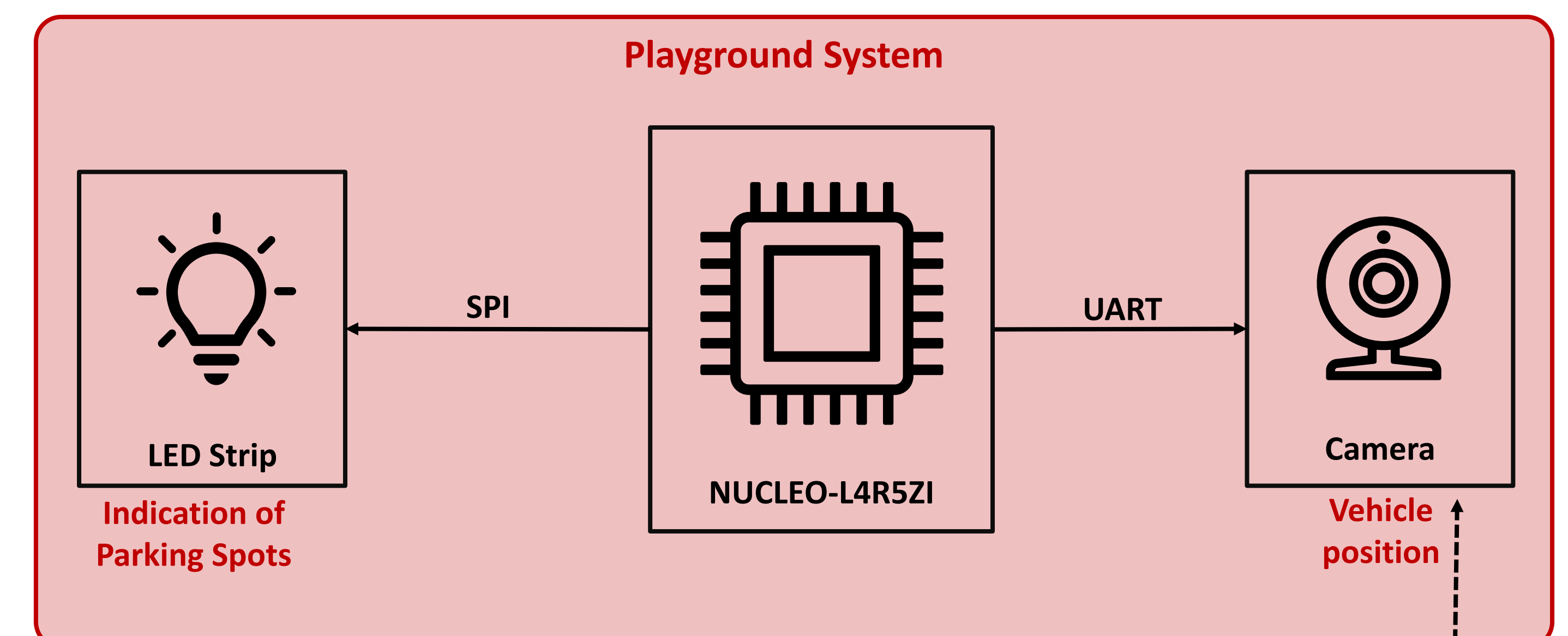
- PixyCam** identifies each Zumo car's position via unique top-mounted color tags.
- Compare available spots position and Zumo's positions to check occupancy.
- Change **LED color** to represent occupancy.

Remote Controller:

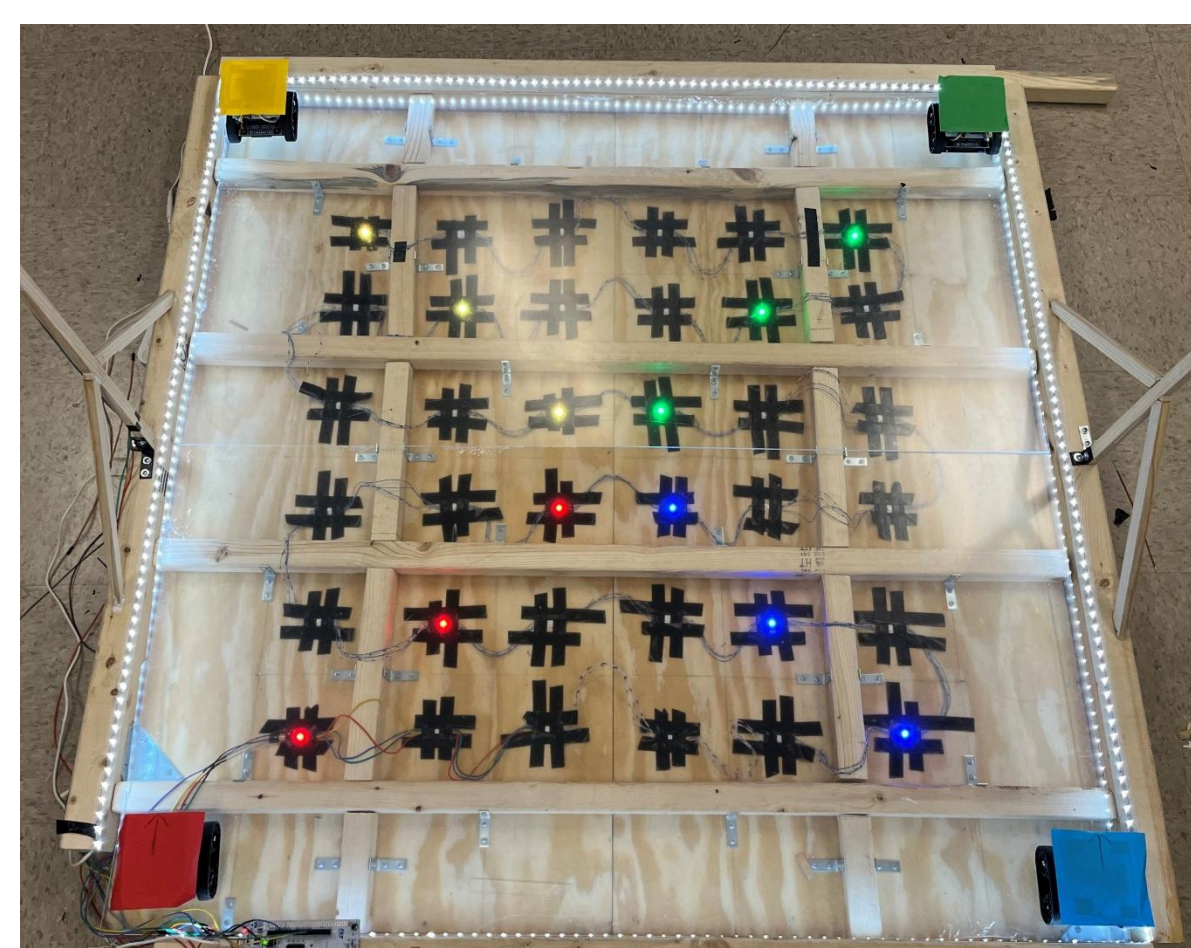
- Reads IMU pitch and roll.
- Compute target speed and direction.
- Sends control signals.
- Force sensitive resistor (FSR)** triggers emergency stop.
- Vibrator** generates haptic feedback.

Zumo Car:

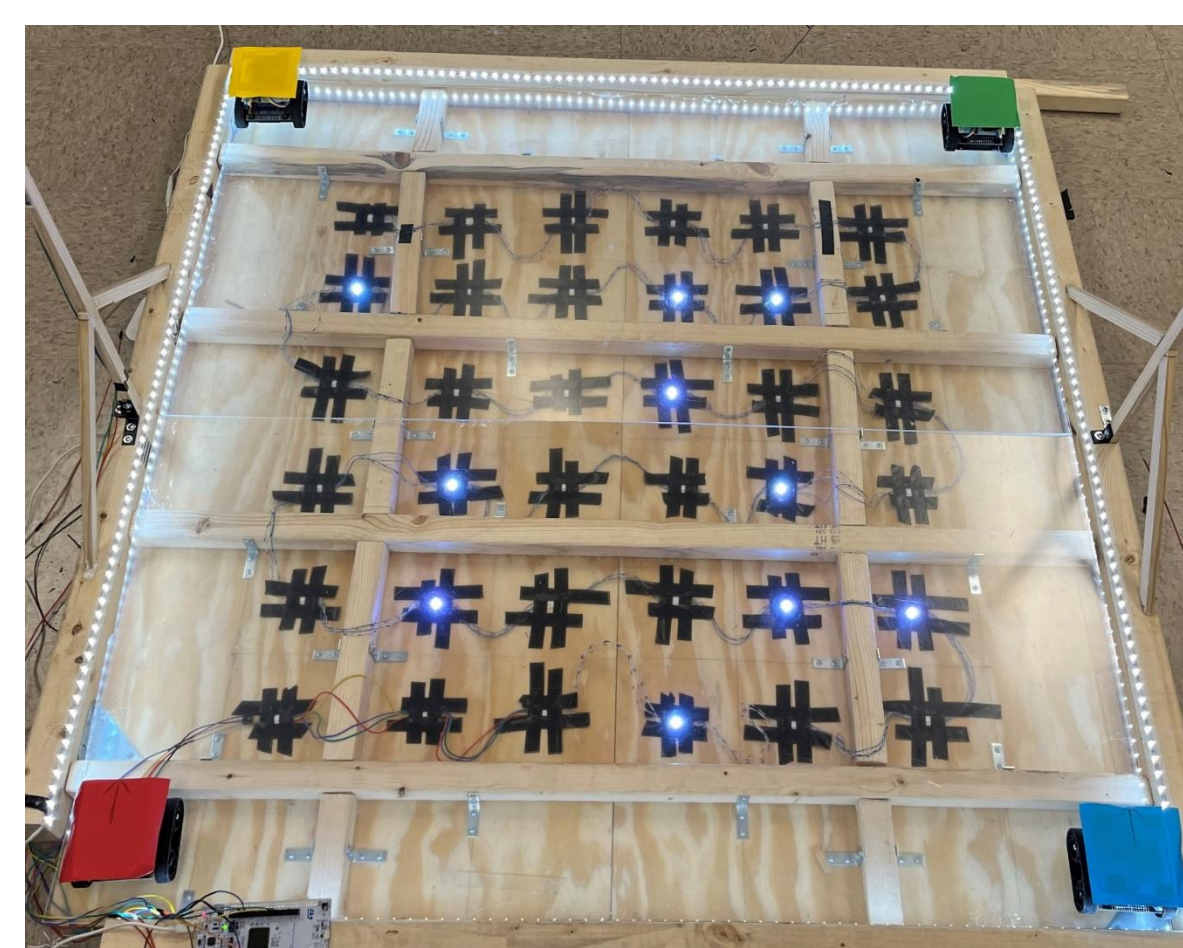
- Reads IMU yaw.
- Receive speed and direction.
- Manipulate the Car using a **PID curve control**.
- Sends velocity feedback.



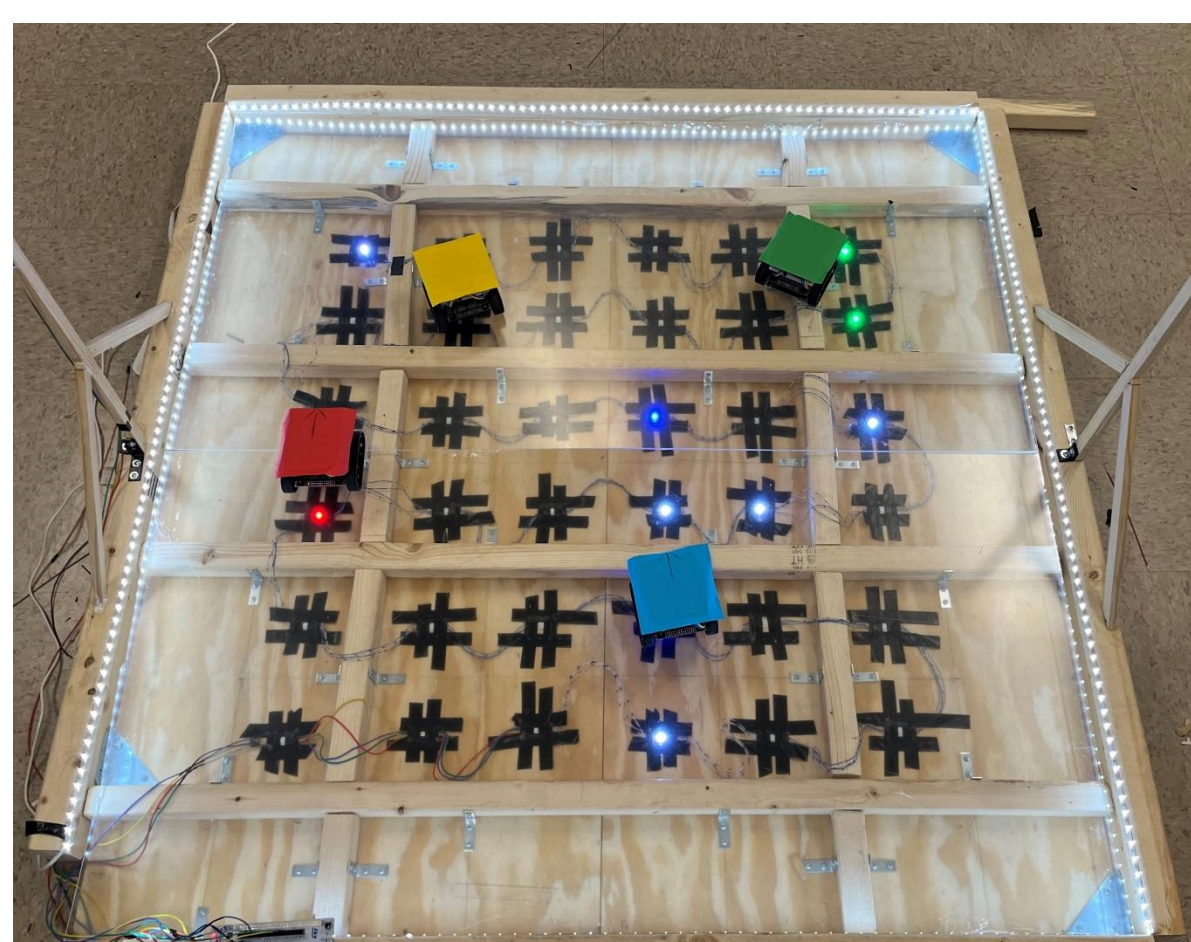
Results



Initialization



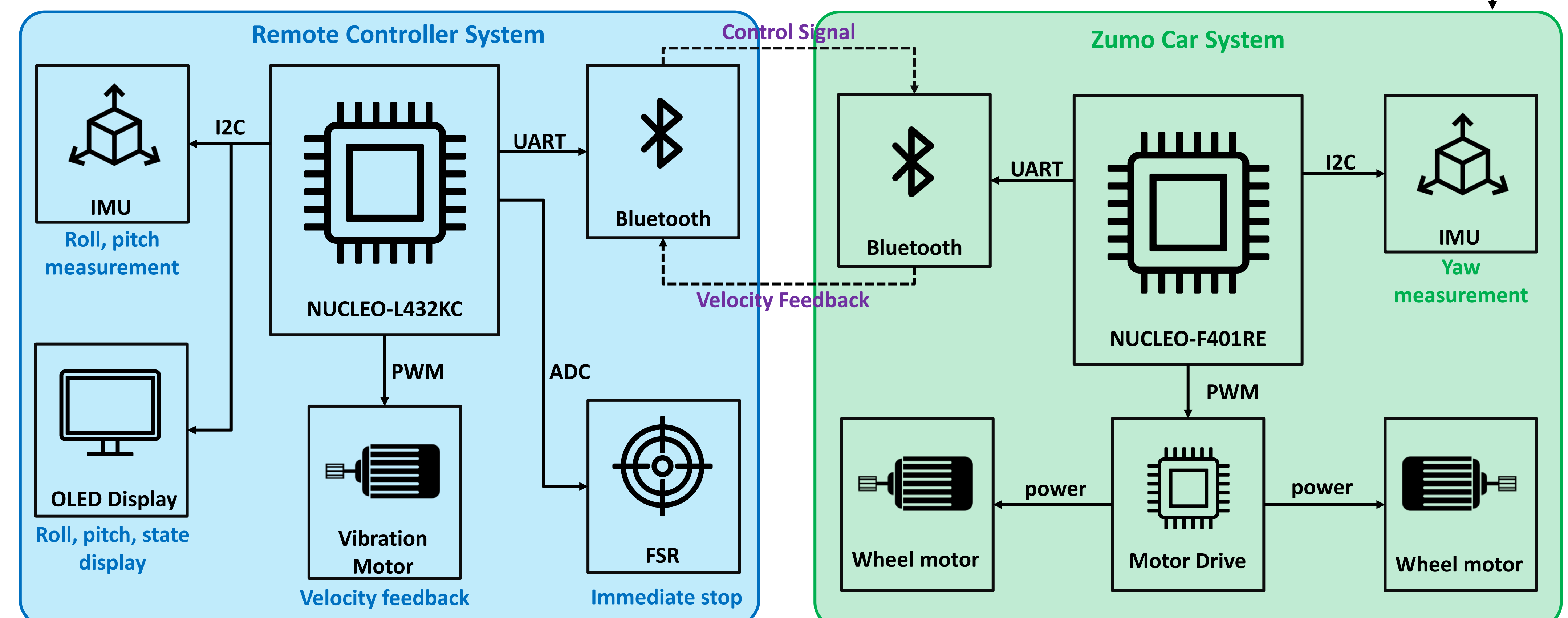
Randomly light up 10 LEDs each round



Zumo Cars compete to occupy spots



Declare winner



Three interconnected systems controlled by STM32 NUCLEO development board

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